File permissions in Linux

Project description

As a security professional supporting a research team, I reviewed and managed Linux file system permissions to ensure users were granted appropriate access. I examined existing permissions, identified mismatches with required authorization, and modified them to remove unauthorized access. Using commands such as `ls -la` and `chmod`, I verified and enforced proper access controls to help maintain system security and protect sensitive research data.

Check file and directory details

I first verified that the projects directory existed by using the `ls` command to list available directories. After confirming its presence, I navigated to the directory using `cd projects`.

Within the directory, I ran `ls -la` to display detailed information for all files and subdirectories, including hidden items. This command provided the permission settings for each file and directory, represented by a **10-character permission string**, which identifies the access levels for the user, group, and others. The screenshot below shows the result of the command executed.

```
researcher2@62a58f96f90f:~$ ls
projects
researcher2@62a58f96f90f:~$ cd projects/
researcher2@62a58f96f90f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 14:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 15:54 ..
-rw--w---- 1 researcher2 research_team   46 Mar 25 14:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 25 14:51 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 25 14:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 25 14:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_t.txt
researcher2@62a58f96f90f:~/projects$
```

Describe the permissions string

Linux file permissions are displayed as a **10-character string**, such as `drwxrwxrwx`, which defines the access level for different users.

- **First character** – Identifies the file type. A `d` represents a directory, while a `-` indicates a regular file. Other symbols may appear for special file types, such as links.
- **Characters 2–4** – Represent the permissions assigned to the **owner (user)**. These include read (`r`), write (`w`), and execute (`x`). A hyphen (`-`) in any position means that specific permission is not granted.
- **Characters 5–7** – Define the permissions for the **group**. These follow the same read, write, and execute structure and determine what members of the file's group are allowed to do.
- **Characters 8–10** – Specify permissions for **others**, which include all users who are not the owner and not part of the group. These permissions control general access to the file or directory.

In addition to the permission string, the expanded directory listing also displays ownership details. The **second column** shows the user who owns the file, while the **third column** identifies the group that owns it. Understanding both permissions and ownership is important for properly managing access control in Linux systems.

Change file permissions

To strengthen system security, I reviewed file permissions to identify any unauthorized access. Specifically, I checked whether any files in the projects directory allowed **others** to have write permissions.

Using the `ls -la` command, I examined the permission settings and discovered that project_k.txt granted write access to others, which posed a security risk. To remove this unauthorized permission, I executed: `chmod o-w project_k.txt`

This command removed the write permission for others while keeping the existing permissions for the user and group unchanged. I then ran `ls -la` again to verify that the changes were successfully applied and that write access for others had been removed. The screenshot below shows the result of the command executed.

```
researcher2@62a58f96f90f:~/projects$ chmod o-w project_k.txt
researcher2@62a58f96f90f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 14:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 15:54 ..
-rw--w---- 1 researcher2 research_team   46 Mar 25 14:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 25 14:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 25 14:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_t.txt
researcher2@62a58f96f90f:~/projects$
```

The file `project_m.txt` is considered a restricted file and should only be accessible by the user. The group and others should not have read or write permissions.

I reviewed the current permissions using `ls -la` and observed that the group had read access to `project_m.txt`, which did not align with the required authorization. To remove this incorrect permission, I executed: `chmod g-r project_m.txt`

This command removed the read permission for the group. I then ran `ls -la` again to verify that the permission change was successfully applied and that only the user retained access to the file. The screenshot below shows the result of the command executed.

```
researcher2@62a58f96f90f:~/projects$ chmod g-r project_m.txt
researcher2@62a58f96f90f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 14:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 15:54 ..
-rw--w---- 1 researcher2 research_team   46 Mar 25 14:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 25 14:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 25 14:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_t.txt
researcher2@62a58f96f90f:~/projects$
```

Change file permissions on a hidden file

To further reduce unauthorized access, I reviewed the permissions of hidden files within the directory. The file .project_x.txt is an archived hidden file and should not be modified. However, both the user and group should retain read access.

Using `ls -la`, I observed that both the user and group had write permissions, which did not align with the required access controls. To correct this, I executed: `chmod u-w,g-w,g+r .project_x.txt`

This command removed write permissions for both the user and group while ensuring read access remained intact. I then ran `ls -la` again to verify that the changes were successfully applied and that the file permissions reflected the intended restrictions. The screenshot below shows the result of the command executed.

```
researcher2@62a58f96f90f:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@62a58f96f90f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 14:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 15:54 ..
-r--r----- 1 researcher2 research_team   46 Mar 25 14:51 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 25 14:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 25 14:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_t.txt
researcher2@62a58f96f90f:~/projects$
```

Change directory permissions

In this task, I reviewed the permissions of the drafts subdirectory within /home/researcher2/projects. Only the **researcher2** user should have access to this directory, meaning that only the user should retain execute privileges.

After checking the current permissions using `ls -la`, I observed that the group had execute access to the drafts directory, which was not authorized. To restrict this access, I executed: `chmod g-x drafts/`

This command removed the execute permission from the group. I then ran `ls -la` again to verify that the change was successfully applied and that only the authorized user retained access to the directory. The screenshot below shows the result of the command executed.

```
researcher2@62a58f96f90f:~/projects$ chmod g-x drafts/
researcher2@62a58f96f90f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 14:51 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 25 15:54 ..
-r--r----- 1 researcher2 research_team   46 Mar 25 14:51 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar 25 14:51 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_k.txt
-rw------- 1 researcher2 research_team   46 Mar 25 14:51 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 25 14:51 project_t.txt
researcher2@62a58f96f90f:~/projects$
```

Summary

I reviewed and corrected Linux file and directory permissions to ensure proper authorization and prevent unauthorized access. Using `ls -la`, I examined existing permissions, including hidden files, and interpreted the 10-character permission string to identify misconfigured access levels.

I applied `chmod` to remove unnecessary write and execute permissions, restrict access to sensitive files, secure hidden archived content, and ensure only authorized users could access critical directories. Each modification was validated by rechecking permissions to confirm the changes.

These actions strengthened access control, reduced security risks, and ensured that files and directories followed the principle of least privilege.